

HEET SONI

AI Engineer | Software Engineer | Full Stack AI Developer

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PROFESSIONAL SUMMARY

AI & Data Science undergraduate with hands-on experience building AI agents, LLM applications, Retrieval-Augmented Generation (RAG) systems, Computer Vision solutions, and Full-Stack Software. Developed end-to-end AI applications through academic, internship, and personal projects, with a focus on scalable backend systems and practical AI solutions.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, C++

AI Technologies: LLMs, AI Agents, Retrieval-Augmented Generation(RAG), Computer Vision, Natural Language Processing(NLP), Prompt Engineering

AI Frameworks & Libraries: TensorFlow, PyTorch, Scikit-learn, OpenCV, YOLOv8

Backend & APIs: FastAPI, Node.js, Express.js, REST APIs, WebSockets

Frontend: React.js, Next.js, Tailwind CSS, HTML5, CSS3

Databases: PostgreSQL, MongoDB, MySQL, ChromaDB, SQLite

Developer Tools: Git, GitHub, Docker, Linux

EDUCATION

B.Tech in Artificial Intelligence and Data Science | A. D. Patel Institute of Technology (ADIT), CVM University, Anand, Gujarat, India

• CGPA: 8.05 / 10 (Till 6th Semester)

June 2023 – June 2027(Expected)

PROJECTS

VAJRA — Foundation Language Model Framework | [Python](#) • [PyTorch](#) • [Transformer Architecture](#) • [Distributed Trainings](#) • [LLMs](#) May 2026 - Present

- Built an open-source foundation language model framework for training, evaluating, and packaging decoder-only Transformer models from scratch with reproducible production pipelines.
- Engineered a modular training and data pipeline supporting custom BPE tokenization, dataset sharding, distributed pretraining, mixed-precision training, checkpointing, and automated evaluation.
- Implemented LLaMA-style Transformer components including RoPE, RMSNorm, SwiGLU, Grouped-Query Attention, agent memory, RAG context building, benchmarking, and Hugging Face-compatible model packaging.

Smart Stadium Copilot | [Python](#) • [FastAPI](#) • [NVIDIA NIM](#) • [NumPy](#) • [Pydantic](#) • [HTML5](#) • [CSS3](#) • [JavaScript](#) • [Pytest](#)

June 2026 – July 2026

- Developed a context-adaptive GenAI copilot that dynamically switches personas, knowledge bases, tool access, and multilingual responses for fans, volunteers, and stadium operators.
- Engineered a secure RAG pipeline with function calling, role-based tool access, prompt-injection protection, PII redaction, and vector-based knowledge retrieval.
- Built a production-style FastAPI backend with 10+ AI tools, comprehensive automated testing (92+ unit tests), and WCAG 2.1 AA accessibility support.

TrafficIQ – Autonomous Multi-Agent Traffic Intelligence System | [Python](#) • [FastAPI](#) • [YOLOv8](#) • [ByteTrack](#) • [ZeroMQ](#) • [SQLAlchemy](#) • [WebSockets](#)

May 2026 – July 2026

- Developed a fully offline edge-native traffic intelligence platform using computer vision and multi-agent coordination to replace fixed-timer traffic signals with adaptive real-time control.
- Engineered a distributed FastAPI system integrating YOLOv8, ByteTrack, ZeroMQ, WebSockets, and deterministic traffic algorithms for real-time vehicle tracking, emergency routing, and live traffic analytics.

AutoPitch Studio | [Python](#) • [FastAPI](#) • [Whisper](#) • [Ollama](#) • [Groq API](#) • [Celery](#) • [Redis](#) • [Node.js](#) • [PptxGenJS](#)

July 2026 – Aug 2026

- Developed an AI-powered presentation generation platform that converts spoken ideas into fully editable Microsoft PowerPoint decks using local speech recognition and LLM-driven content generation.
- Engineered a modular pipeline integrating Whisper, Ollama/Groq, Celery, Redis, and PptxGenJS to support privacy-first processing, asynchronous task execution, and native PPTX generation.

WORK EXPERIENCE

Artificial Intelligence Intern | [Codec Technologies](#) | [Python](#) • [NumPy](#) • [Scikit-learn](#) • [TensorFlow](#) • [NLP](#) • [Machine Learning](#)

May 2026 – June 2026

- Developed machine learning and deep learning applications, including a Multilayer Perceptron (MLP) implemented from scratch using NumPy for handwritten digit recognition and an NLP-based spam email classifier using TF-IDF and Support Vector Machines.
- Built interactive AI applications by visualizing neural network activations, evaluating multiple classification models, and applying feature engineering, model training, and performance optimization on real-world datasets.

AI-Assisted Full Stack Web Development | [Graphic Era Deemed to be University](#) | [AI-Assisted Development](#) • [Full-Stack Development](#) • [Debugging](#) • [Software Engineering](#)

June 2026 – August 2026

- Developed AI-assisted full-stack web solutions, applying modern AI tools for code generation, debugging, optimization, and rapid problem solving across frontend and backend development.
- Built and improved web-based applications by combining AI-assisted workflows with conventional programming, translating technical requirements into functional solutions and resolving development challenges.

CERTIFICATIONS

- Google Machine Learning Crash Course — [Google](#)
- Build Your Own Small Language Model — [Google DeepMind](#)
- Hewlett Packard Enterprise – Software Engineering Job Simulation — [Forage](#)
- Datacom – Automation AI Accelerator: From Co-pilot to Autonomous Agent — [Forage](#)
- Vibe2Ship – India's Biggest Vibe Coding Hackathon — [Coding Ninjas](#)
- BCG – GenAI Job Simulation — [Forage](#)

TECHNICAL COMPETITIONS

- **PromptWars Virtual (Hack2Skill)** — Ranked #86 / 43,260 (Top 0.2%) in the AI Code Submission Challenge.
- **Google's AI Agents: Intensive Vibe Coding Capstone** — Built and deployed Agentic AI solutions using modern LLM workflows.
- **DoraHacks Casper Agentic Buildathon** — Developed AI agent applications in a competitive hackathon environment.
- **Skill-to-Agent Dual Cascade Hackathon (DoraHacks)** — Developed reusable AI agent skills and autonomous agent workflows in a competitive global hackathon focused on the AI Agent ecosystem.
- **Vibe2Ship 2026 (Coding Ninjas)** — Participated in India's largest Vibe Coding Hackathon.